SmartRoute DAA PROJECT

Emergency Medical Response & Dynamic Dispatch System

Comprehensive System Report
Date: October 2026
Status: Active & Deployed

1. Executive Summary

SmartRoute is a full-stack emergency medical response dispatch application designed to drastically minimize Emergency Response Times (ETA). By leveraging core computer science principles—specifically **Design and Analysis of Algorithms (DAA)** graph traversal (Dijkstra's Shortest Path) with dynamic live rerouting—the system coordinates patients, ambulances, and hospital trauma facilities in real-time.

🎯 Core Project Objectives

- **Immediate Dispatch Without Sign-up:** Patients can trigger emergency requests in seconds without entering credentials or downloading apps.
- **Condition-Driven Medical Kit Recommendation:** Dynamic triage algorithm suggests the appropriate onboard emergency medical kit (e.g., cardiac defibrillators, burn packs, fracture kits).
- **Dynamic Patient Hospital Selection & Rerouting:** Allows patients or guardians to select their preferred hospital, automatically recalculating the ambulance's route in real time.
- **Hybrid Backend Storage:** Dual-mode architecture utilizing **Supabase (PostgreSQL)** with automatic offline fallback to **Browser LocalStorage**.

2. Technology Stack & Architecture

Layer	Technologies Used	Architectural Function
Frontend Interface	HTML5, Vanilla CSS3 (Glassmorphism), ES6+ JavaScript	Zero framework overhead, hyper-responsive UI with sub-second rendering across mobile and desktop.
DAA Visualizer	HTML5 Canvas 2D API & Animation Loop	Renders weighted road network graphs, node relaxation, and live path rerouting visualization.
Web Server	Node.js `http` Native Module (Port 3000)	Clean URL routing, static file streaming, mime-type sanitization, and live request logging.
Primary Database	Supabase Cloud (PostgreSQL + PostGIS)	Relational incident tracking, hospital ER capacity, and ambulance unit telemetry.
Fallback Storage	HTML5 Web Storage (`localStorage`)	Guarantees 100% functionality offline and during live academic/classroom evaluations.

3. System Modules & Features

🚨 1. User Emergency Portal

`user-portal.html`

- **One-Tap Geolocation:** Automatically fetches coordinates using the HTML5 Geolocation API.
- **Triage & Severity Assessment:** Captures condition, severity (Critical, High, Medium, Low), and contact info.
- **Smart Kit Matching:** Recommends kit configurations based on selected medical conditions.
- **Hospital Choice:** Patients pick from nearby hospitals with real-time bed count display.
- **Live Ticket Status:** Real-time ETA countdown, unit code, and driver contact.

⚡ 2. DAA Algorithm Visualizer

`daa-explain.html`

- **Weighted Graph Canvas:** Visualizes Ambulance Base (A), Patient (P), Junctions (J1-J3), and Hospitals (h1-h4).
- **Dijkstra Step-by-Step:** Interactive step relaxation showing visited nodes, tentative distances, and queue state.
- **Dynamic Rerouting:** Demonstrates real-time recalculation when a destination hospital is switched mid-journey.
- **Time/Space Complexity:** Clear DAA analysis cards comparing Dijkstra, A*, and Floyd-Warshall.

🛡️ 3. Admin Control Panel

`admin-dashboard.html`

- **Real-Time Incident Stream:** Live feeds of active dispatches, priorities, and statuses.
- **Hospital Bed Manager:** Instant updating of ER beds, ICU availability, and status.
- **Fleet Monitor:** Track status of ambulances (Available, Busy, Maintenance).
- **Database Settings:** Configure live Supabase URL and keys directly from UI.

🌐 4. Landing Page

`index.html`

- **Live Dispatch Simulation:** Animated radar map demonstrating ambulance-to-patient tracking.
- **Quick Access Hub:** Instant navigation links to User, Admin, and Algorithm modules.
- **Service Metrics:** Displays network metrics, coverage stats, and average response times.

4. Algorithmic Foundation (DAA Implementation)

The dispatch routing engine models the emergency transportation network as a directed weighted graph `G = (V, E)`, where vertices represent locations and edges represent roads with dynamic weight factors.

```
Edge Weight W(u, v) = Distance(u, v) × Traffic_Multiplier(u, v)
```

To find the optimal trajectory from the dispatched ambulance unit through the patient location to the selected hospital, Dijkstra's algorithm is executed:

```
// Core Relaxation Step in DAA Visualizer
for (const neighbor of graph[currNode]) {
    const edgeWeight = neighbor.dist * neighbor.traffic;
    const altDistance = dist[currNode] + edgeWeight;

    if (altDistance < dist[neighbor.to]) {
        dist[neighbor.to] = altDistance;
        prev[neighbor.to] = currNode;
        priorityQueue.insert({ node: neighbor.to, priority: altDistance });
    }
}
```

Algorithm	Time Complexity	Space Complexity	Role in SmartRoute
Dijkstra (Min-Heap)	`O((V + E) log V)`	`O(V)`	Primary path calculation from patient to nearest available trauma centers.
A* Search (Heuristic)	`O(E)` (Best) / `O(b^d)`	`O(V)`	Direct ambulance dispatch when GPS coordinates provide Euclidean distance heuristics.
Dynamic Rerouting	`O(V log V)`	`O(V)`	Re-plans path from current ambulance position upon patient hospital switch.

5. Backend Connection & Database Architecture

SmartRoute features a **dual-mode abstraction layer** implemented in `scripts/supabase-client.js`. The application automatically determines whether to send queries to a live cloud Supabase instance or save locally.

Database Relational Model (PostgreSQL / Supabase)

```
-- 1. HOSPITALS TABLE
CREATE TABLE hospitals (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    name VARCHAR(255) NOT NULL,
    latitude NUMERIC(10, 6) NOT NULL,
    longitude NUMERIC(10, 6) NOT NULL,
    available_er_beds INT DEFAULT 10,
    icu_available BOOLEAN DEFAULT TRUE,
    hospital_type VARCHAR(100),
    status VARCHAR(50) DEFAULT 'active'
);

-- 2. AMBULANCES TABLE
CREATE TABLE ambulances (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    unit_code VARCHAR(50) UNIQUE NOT NULL,
    driver_name VARCHAR(100) NOT NULL,
    driver_phone VARCHAR(50),
    current_status VARCHAR(50) DEFAULT 'Available',
    kit_loaded TEXT
);

-- 3. INCIDENTS TABLE
CREATE TABLE incidents (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    ticket_code VARCHAR(50) UNIQUE NOT NULL,
    patient_name VARCHAR(150),
    emergency_type VARCHAR(100) NOT NULL,
    severity_level VARCHAR(20) NOT NULL,
    selected_hospital_id UUID REFERENCES hospitals(id),
    assigned_ambulance_id UUID REFERENCES ambulances(id),
    recommended_kit JSONB,
    status VARCHAR(50) DEFAULT 'Dispatched',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

6. Running & Verification

🚀 Local Deployment Guide

1. Run server: `node server.js`
2. Open Landing Page: **http://localhost:3000/**
3. Open User Portal: **http://localhost:3000/user-portal.html**
4. Open DAA Visualizer: **http://localhost:3000/daa-explain.html**
5. Open Admin Dashboard: **http://localhost:3000/admin-dashboard.html** (Admin Login: `admin@smartroute.com` / `admin123`)